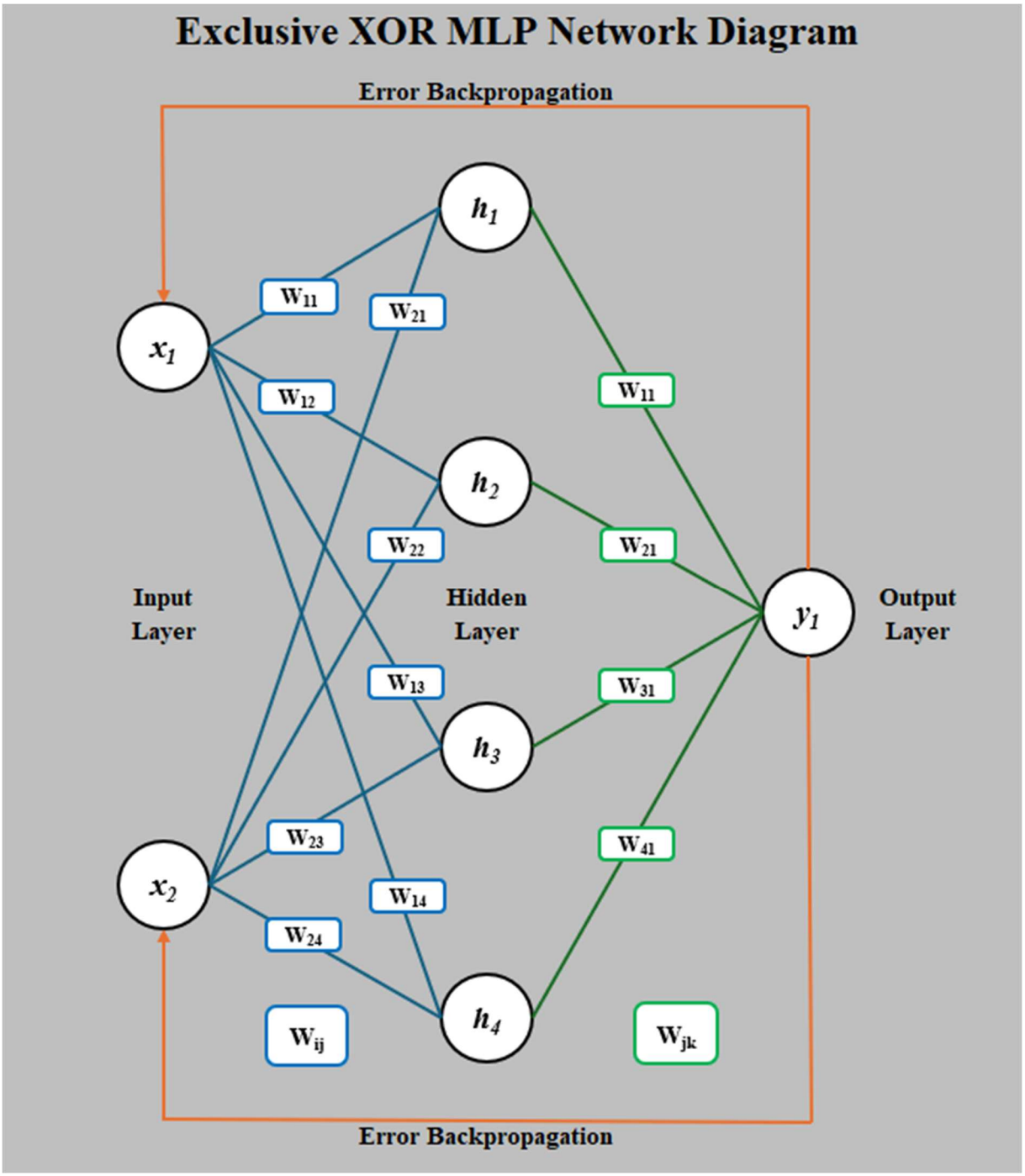
```
import numpy as np
import tensorflow as tf
from tensorflow import keras
import matplotlib.pyplot as plt

# Set a seed for reproducibility.
tf.random.set_seed(42)

# Data matching I/O table.
X = np.array([[0, 0], [0, 1], [1, 0], [1, 1]])
y = np.array([[0], [1], [1], [0]])
```

Exclusive XOR I/O Table and MLP Network Diagram will be made available in the complementary .pdf file.

Exclusive XOR I/O Table

Inputs		output
x_1	x_2	y
1	1	0
0	1	1
1	0	1
0	0	0




```python
# Model with tanh activation for better gradients.
model = keras.Sequential([
    keras.layers.Dense(4, input_dim=2, activation='tanh'),  # Hidden layer.
    keras.layers.Dense(1, activation='sigmoid')
])
```

```python
# Compile with Backpropagation learning rate and MSE loss for chart
consistency.
model.compile(optimizer=keras.optimizers.Adam(learning_rate=0.1),
              loss='mean_squared_error',
              metrics=['accuracy'])

# Define model training length.
history = model.fit(X, y, epochs=200, verbose=1)

# Print epochs used.
print(f"Stopped after {len(history.history['loss'])} epochs")
```

```
Epoch 1/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 627ms/step - accuracy: 0.5000 - loss: 0.2722
Epoch 2/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 0.5000 - loss: 0.2597
Epoch 3/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 0.5000 - loss: 0.2564
Epoch 4/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 0.5000 - loss: 0.2534
Epoch 5/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 0.5000 - loss: 0.2490
Epoch 6/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 0.5000 - loss: 0.2443
Epoch 7/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 0.5000 - loss: 0.2405
Epoch 8/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 0.7500 - loss: 0.2368
Epoch 9/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 0.7500 - loss: 0.2321
Epoch 10/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 0.7500 - loss: 0.2254
Epoch 11/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 0.7500 - loss: 0.2169
Epoch 12/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 0.5000 - loss: 0.2073
Epoch 13/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 0.5000 - loss: 0.1971
Epoch 14/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 0.7500 - loss: 0.1855
Epoch 15/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.1721
Epoch 16/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 0.7500 - loss: 0.1580
Epoch 17/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 0.7500 - loss: 0.1450
Epoch 18/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 0.7500 - loss: 0.1315
Epoch 19/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.1167
```

```
Epoch 20/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.1022
Epoch 21/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0891
Epoch 22/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.0770
Epoch 23/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0653
Epoch 24/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0548
Epoch 25/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.0461
Epoch 26/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0386
Epoch 27/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.0320
Epoch 28/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 63ms/step - accuracy: 1.0000 - loss: 0.0263
Epoch 29/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 53ms/step - accuracy: 1.0000 - loss: 0.0218
Epoch 30/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0182
Epoch 31/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 0.0153
Epoch 32/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0129
Epoch 33/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.0109
Epoch 34/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.0093
Epoch 35/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.0079
Epoch 36/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0068
Epoch 37/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.0059
Epoch 38/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 0.0052
Epoch 39/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.0046
Epoch 40/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.0041
Epoch 41/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0037
Epoch 42/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.0033
Epoch 43/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - accuracy: 1.0000 - loss: 0.0030
Epoch 44/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 0.0028
```

```
Epoch 45/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0025
Epoch 46/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - accuracy: 1.0000 - loss: 0.0023
Epoch 47/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0022
Epoch 48/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 0.0020
Epoch 49/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0019
Epoch 50/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 1.0000 - loss: 0.0017
Epoch 51/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0016
Epoch 52/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 0.0016
Epoch 53/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.0015
Epoch 54/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.0014
Epoch 55/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0013
Epoch 56/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.0013
Epoch 57/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0012
Epoch 58/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 0.0012
Epoch 59/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 0.0011
Epoch 60/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0011
Epoch 61/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0011
Epoch 62/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 0.0010
Epoch 63/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 1.0000 - loss: 9.9850e-04
Epoch 64/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 9.7054e-04
Epoch 65/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 9.4463e-04
Epoch 66/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 9.2055e-04
Epoch 67/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 8.9812e-04
Epoch 68/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 8.7718e-04
Epoch 69/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 1.0000 - loss: 8.5759e-04
```

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Epoch 70/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 8.3921e-04
Epoch 71/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 8.2194e-04
Epoch 72/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 8.0567e-04
Epoch 73/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 7.9032e-04
Epoch 74/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 7.7580e-04
Epoch 75/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 7.6204e-04
Epoch 76/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 7.4898e-04
Epoch 77/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 7.3655e-04
Epoch 78/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 7.2469e-04
Epoch 79/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 7.1337e-04
Epoch 80/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 7.0252e-04
Epoch 81/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 6.9212e-04
Epoch 82/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 6.8213e-04
Epoch 83/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 6.7251e-04
Epoch 84/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 6.6323e-04
Epoch 85/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 6.5428e-04
Epoch 86/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 6.4561e-04
Epoch 87/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 6.3721e-04
Epoch 88/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 6.2907e-04
Epoch 89/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 6.2116e-04
Epoch 90/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 6.1347e-04
Epoch 91/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 6.0599e-04
Epoch 92/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 5.9870e-04
Epoch 93/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 5.9159e-04
Epoch 94/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - accuracy: 1.0000 - loss: 5.8466e-04
```

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Epoch 95/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 1.0000 - loss: 5.7788e-04
Epoch 96/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 5.7126e-04
Epoch 97/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 5.6479e-04
Epoch 98/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 89ms/step - accuracy: 1.0000 - loss: 5.5845e-04
Epoch 99/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 5.5224e-04
Epoch 100/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 5.4617e-04
Epoch 101/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 5.4021e-04
Epoch 102/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 5.3437e-04
Epoch 103/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 5.2863e-04
Epoch 104/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 5.2301e-04
Epoch 105/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 5.1749e-04
Epoch 106/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 5.1206e-04
Epoch 107/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 5.0673e-04
Epoch 108/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 5.0149e-04
Epoch 109/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 4.9634e-04
Epoch 110/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 4.9127e-04
Epoch 111/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 4.8629e-04
Epoch 112/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 4.8139e-04
Epoch 113/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 4.7656e-04
Epoch 114/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 4.7181e-04
Epoch 115/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 4.6713e-04
Epoch 116/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 4.6252e-04
Epoch 117/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 1.0000 - loss: 4.5799e-04
Epoch 118/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 4.5352e-04
Epoch 119/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 4.4911e-04
```

```
Epoch 120/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 4.4477e-04
Epoch 121/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 4.4049e-04
Epoch 122/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 4.3628e-04
Epoch 123/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 4.3212e-04
Epoch 124/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 4.2802e-04
Epoch 125/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 4.2398e-04
Epoch 126/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 4.2000e-04
Epoch 127/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 4.1607e-04
Epoch 128/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 4.1219e-04
Epoch 129/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 4.0837e-04
Epoch 130/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 1.0000 - loss: 4.0460e-04
Epoch 131/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 4.0088e-04
Epoch 132/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.9722e-04
Epoch 133/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 1.0000 - loss: 3.9360e-04
Epoch 134/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.9003e-04
Epoch 135/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.8650e-04
Epoch 136/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.8302e-04
Epoch 137/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.7959e-04
Epoch 138/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.7621e-04
Epoch 139/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 3.7286e-04
Epoch 140/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 3.6957e-04
Epoch 141/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 3.6631e-04
Epoch 142/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 3.6309e-04
Epoch 143/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.5992e-04
Epoch 144/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 1.0000 - loss: 3.5679e-04
```

```
Epoch 145/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.5370e-04
Epoch 146/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 3.5065e-04
Epoch 147/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.4763e-04
Epoch 148/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 3.4466e-04
Epoch 149/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.4172e-04
Epoch 150/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.3882e-04
Epoch 151/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 83ms/step - accuracy: 1.0000 - loss: 3.3595e-04
Epoch 152/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 1.0000 - loss: 3.3312e-04
Epoch 153/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.3033e-04
Epoch 154/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.2757e-04
Epoch 155/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.2484e-04
Epoch 156/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 3.2215e-04
Epoch 157/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 3.1949e-04
Epoch 158/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 3.1686e-04
Epoch 159/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 3.1427e-04
Epoch 160/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.1171e-04
Epoch 161/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 1.0000 - loss: 3.0918e-04
Epoch 162/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 3.0667e-04
Epoch 163/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 3.0420e-04
Epoch 164/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 3.0176e-04
Epoch 165/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 2.9935e-04
Epoch 166/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 2.9697e-04
Epoch 167/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 1.0000 - loss: 2.9461e-04
Epoch 168/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 2.9228e-04
Epoch 169/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 2.8998e-04
```

```
Epoch 170/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 2.8771e-04
Epoch 171/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 2.8547e-04
Epoch 172/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 2.8325e-04
Epoch 173/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 2.8105e-04
Epoch 174/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 2.7888e-04
Epoch 175/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 2.7674e-04
Epoch 176/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 2.7462e-04
Epoch 177/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 2.7253e-04
Epoch 178/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 2.7046e-04
Epoch 179/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 2.6841e-04
Epoch 180/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 2.6639e-04
Epoch 181/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 1.0000 - loss: 2.6439e-04
Epoch 182/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 2.6241e-04
Epoch 183/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 2.6046e-04
Epoch 184/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 2.5853e-04
Epoch 185/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 2.5661e-04
Epoch 186/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 2.5473e-04
Epoch 187/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 2.5286e-04
Epoch 188/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 2.5101e-04
Epoch 189/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 2.4918e-04
Epoch 190/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 2.4738e-04
Epoch 191/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 2.4559e-04
Epoch 192/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 2.4382e-04
Epoch 193/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 2.4207e-04
Epoch 194/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 2.4034e-04
```

```
Epoch 195/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - accuracy: 1.0000 - loss: 2.3863e-04
Epoch 196/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 2.3694e-04
Epoch 197/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - accuracy: 1.0000 - loss: 2.3527e-04
Epoch 198/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - accuracy: 1.0000 - loss: 2.3361e-04
Epoch 199/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 83ms/step - accuracy: 1.0000 - loss: 2.3198e-04
Epoch 200/200
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - accuracy: 1.0000 - loss: 2.3036e-04
Stopped after 200 epochs
```

Hidden layer densities of 2, 4, 6, and 8 were tested along with Sigmoid, tanh, and ReLU activation functions. A hidden layer density of 4 and an activation function of tanh seemed to generate the best results. 0.01, 0.1, and 0.2 Backpropogation learning rates the Adam optimizer were tested, and the 0.1 rate achieved the best results. Chose to use 'mean_squared_error' for a loss method, instead of 'binary_crossentropy', this seemed better in matching classic backdrops for XOR, where MSE is common and can drop very low with sigmoid activations. An epoch value was targeted in such a way to achieve a Sum of Squared Error result < 0.001, settling on 200. Higher number of epochs improved the performance even further, but setting a benchmark result made sense.

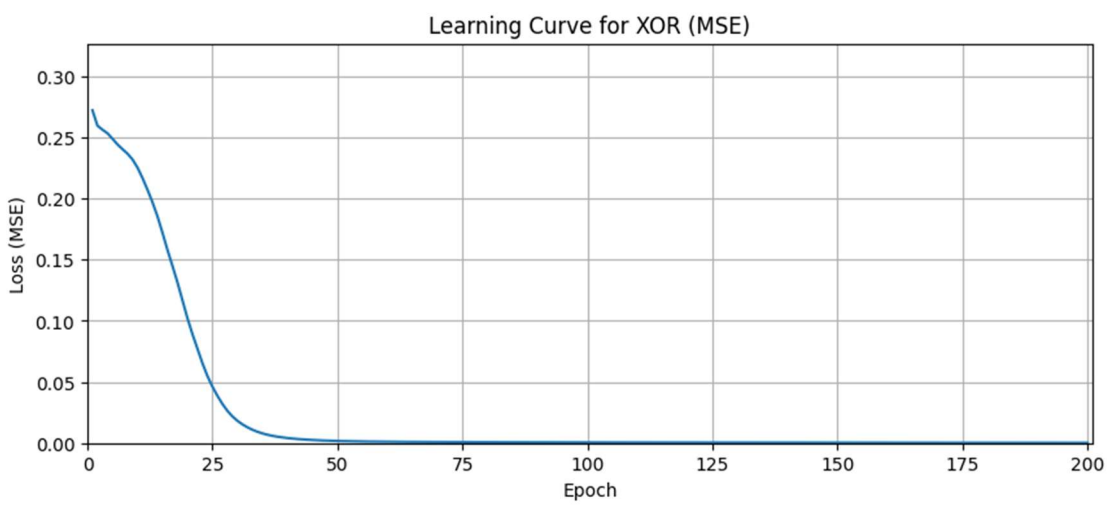
```
# Model Evaluations
loss, accuracy = model.evaluate(X, y, verbose=0)
print(f"\nEvaluation - Loss: {loss:.6f}, Accuracy: {accuracy:.4f}")
```

```
Evaluation - Loss: 0.000229, Accuracy: 1.0000
```

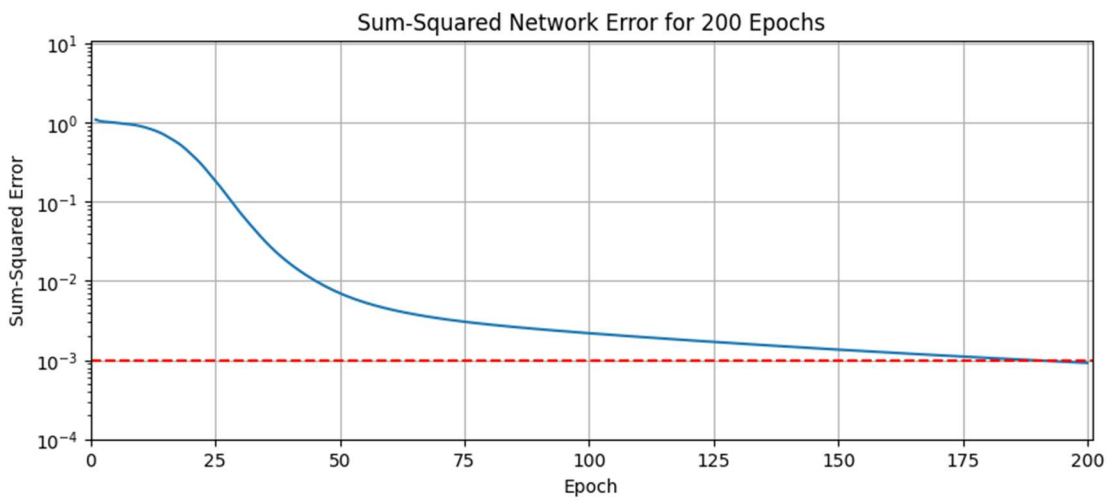
```
# Model Predictions
predictions = model.predict(X)
rounded_predictions = np.round(predictions)

print("\nPredictions:")
squared_errors = []
for i in range(len(X)):
    print(f"Input: {X[i]}, True Output: {y[i][0]}, Predicted:
{rounded_predictions[i][0]}, Raw Prob: {predictions[i][0]:.4f}, Errors:
{rounded_predictions[i][0]-predictions[i][0]:.4f}")
    squared_error = (predictions[i][0]-y[i][0])**2
    squared_errors.append(squared_error)

SSE = np.sum(squared_errors)
print(f"\nSum of Squared Errors: {SSE:.5f}")
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 50ms/step

Predictions:
Input: [0 0], True Output: 0, Predicted: 0.0, Raw Prob: 0.0083, Errors: -
0.0083
Input: [0 1], True Output: 1, Predicted: 1.0, Raw Prob: 0.9823, Errors:
0.0177
Input: [1 0], True Output: 1, Predicted: 1.0, Raw Prob: 0.9860, Errors:
0.0140
Input: [1 1], True Output: 0, Predicted: 0.0, Raw Prob: 0.0184, Errors: -
0.0184

Sum of Squared Errors: 0.00092
```

```python
# Plotting data.
epochs = range(1, len(history.history['loss']) + 1)
losses = np.array(history.history['loss'])
sse = losses * 4  # Sum-squared error

# Linear MSE plot.
plt.figure(figsize=(10, 4))
plt.plot(epochs, losses)
plt.title('Learning Curve for XOR (MSE)')
plt.xlabel('Epoch')
plt.ylabel('Loss (MSE)')
plt.xlim(0, 201)
plt.ylim(0, max(losses[:50]) * 1.2)
plt.grid(True)
plt.show()
```



```
# SSE on log scale plot.
plt.figure(figsize=(10, 4))
plt.plot(epochs, sse)
plt.yscale('log')
plt.title('Sum-Squared Network Error for 200 Epochs')
plt.xlabel('Epoch')
plt.ylabel('Sum-Squared Error')
plt.xlim(0, 201)
plt.ylim(max(1e-4, min(sse)/10), max(sse) * 10)
plt.axhline(y=0.001, color='red', linestyle='--')
plt.grid(True)
plt.show()
```



Though it is not apparent in this program, experimentation with an SGD optimizer and momentum, AdaptiveLR and callbacks, custom stops and early stopping, and a learning rate (LR) adaptation plot took place. In the end, the methods, functions, and parameters used seemed the most straight forward and effective. Needless to say, this assignment was definitely a valuable learning experience.